

CANDIDATE LISTS ARE PRIORS: EVIDENCE-FRAME DEPENDENCE IN ACTIVE MODEL SELECTION

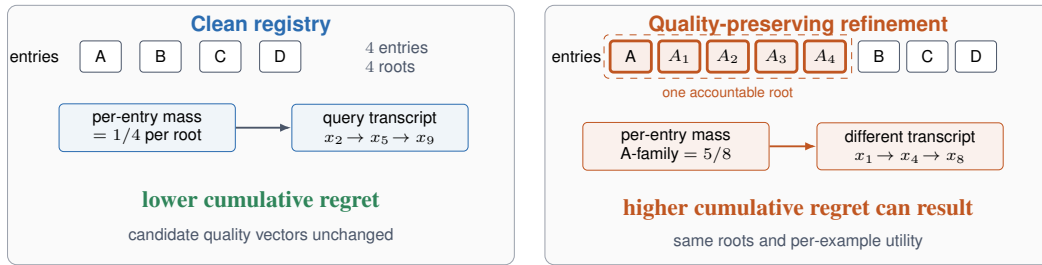
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ABSTRACT

Active model selection uses candidate outputs to decide which costly examples to label, so the supplied list silently defines a prior over the experiment. We show that this representation is not invariant. In frozen responses from 31 models, restoring one natural exact version alias changes all 1,500 paired acquisition paths despite adding no behavior; registry-independent querying is unchanged. We formalize the missing *candidate evidence frame* and prove a tight annotation-budget-linear lower bound for selectors that see responses but not which entries deserve independent mass. In a source-faithful exact-match Select-LLM specialization, our main public-reference intervention preserves correctness coordinate-wise, yet raises cumulative regret by 3.25/3.78/.85 on MedQA/GSM8K/OpenBookQA and terminal regret by 2.91/10.56 points on the first two. Fixed-query effects are zero; 16 permutations and all nine exact tie policies preserve the result. A reference-free decoupled-view test passes terminal and active-minus-fixed gates on 3/8 untouched holdouts; learned raw-input adapters raise AG News terminal regret by .547 [.502, .593] points. Stricter learned single-s confirmations change 99.8/100% of paths and raise Banking77/20 Newsgroups cumulative regret by .0409 [.0129, .0689]/.0822 [.0385, .1266], but both fail frozen terminal gates. An official pairwise-judge selector passes on 2/6 datasets. Exact quotienting misses non-identical aliases and untargeted additions are mostly benign. These results establish targeted evidence-frame dependence, not universal instability or production compromise; active evaluation should make candidate mass and evidence views auditable.

1 INTRODUCTION



Predictions expose submitted response rows, but not the accountable map $\rho : E \rightarrow R$ or which entries deserve independent evidential mass.

Figure 1: The candidate list is an empirical prior. Adding entries from one accountable root can change active queries even when the added entries introduce no new per-example quality vector. The observable responses do not authenticate the root map. This schematic illustrates the mechanism; Tables 3 and 7 report measured effects.

Expert labels are costly, so DiffUse, MODEL SELECTOR, CODA, LLM Selector, and Select-LLM use candidate outputs, weak judges, or consensus to prioritize labels for model comparison (Ashury Tahan et al., 2024; Okanovic et al., 2025; Kay et al., 2025; Durmazkeser et al., 2025; 2026).

In the candidate-indexed Bayesian and consensus mechanisms we audit, however, every listed entry receives separate probability or consensus mass. They state no invariance requirement for two lists encoding the same behavioral alternatives.

This is not only synthetic. In our frozen roster, `claude-instant-1.2` and `claude-instant-v1` have identical responses and scores on all 2,500 task items. Restoring the latter to its 30-entry quotient raises the behavior’s mass from 1/30 to 2/31 and changes all 1,500 Select-LLM paths; random querying remains equal. Unique-label substitution is only 8.7/4.7/3.2% of budget, so this establishes transcript dependence rather than broad natural harm.

The missing object is the *candidate evidence frame*: a map from listed *submitted entries* to accountable *candidate roots*, plus a mass rule. Even with every response, an evaluator may lack the ownership or lineage evidence for this map. Treating entries as the population then makes registry representation an unreported experimental prior.

We next test exploitability after literal copies are removed. T2 uses public or independently acquired references to add response-distinct entries that preserve correctness coordinate-wise while fixing all existing candidates, root qualities, labels, pools, and code. It is a causal stress test, not hidden-label admission. T1 instead learns on calibration and receives no holdout reference or peer output; eight untouched QA/translation scenarios provide the prospective realism bridge. Figure 1 illustrates T2.

Sequential multiplicity is prior in strategic-replication bandits, perturbed-clone active evaluation, redundant-committee quotients, and static leaderboards (Shin et al., 2022; 2025; Lanctot et al., 2026; Nguyen et al., 2025; Singh et al., 2025; Hays et al., 2026). We claim neither the first sequential clone effect nor the first multiplicity attack. Our object is a latent same-root refinement in shared-label model selection: entry mass changes which reusable references are acquired and the resulting root-level regret. Static ranks can be recomputed after quotienting, but an acquisition transcript cannot recover labels never requested. Hashes remove exact copies; similarity cannot establish ownership. Our question complements efficient-evaluation reliability audits (Yauney et al., 2026): which candidates constitute the evidential population?

We prove a tight budget-linear loss and construct four aliases that redirect every query. Exact-quality public-reference interventions raise cumulative regret on three tasks and terminal regret by 2.91/10.56 points on MedQA/GSM8K; candidate order and nine exact tie policies preserve the result. Three of eight prospective reference-free scenarios pass terminal and adaptive-minus-fixed gates. These F1/BLEU tests and a learned AG News success use explicitly decoupled acquisition and scoring views. A stricter learned, no-lookup Banking77 audit uses the *same* exact-match similarity for candidate agreement and reference feedback: cumulative regret rises, but terminal harm is null and the frozen gate fails. An official LLM Selector audit passes corrected terminal gates on two of six datasets; CODA and all negative cells bound the claim to selectable, not generic, degradation.

Our contributions are:

1. We define registry refinement and evidence-frame ambiguity, then show that a natural exact alias rewrites every paired path across three tasks without adding behavior.
2. Executable refinements cause fresh-pool harm. Reference-free decoupled-view tests pass 3/8 untouched holdouts, learned raw-input adapters pass AG News, and a pairwise-judge selector passes 2/6 datasets. A same-similarity Banking77 test changes paths and cumulative cost but fails its terminal gate. Fixed-query effects are zero throughout.
3. We prove a tight budget-linear lower bound and construct four aliases that redirect all B queries and cause $\Omega(B)$ cumulative regret.
4. Exact quotienting misses the main aliases; covers and clone-robust weights trade suppression against clean selection. Untargeted additions are mostly benign.

We do not claim that every active selector is vulnerable, that a natural alias is adversarial, that every refinement is harmful, or that similarity reveals ownership. The resulting reliability requirement is narrower: an active evaluation should either bind submitted entries to a declared candidate evidence frame or qualify conclusions that depend on unaudited entry multiplicity.

2 SETTING, THREAT MODEL, AND CLOSEST BOUNDARIES

Active selection. Let $\mathcal{X} = \{x_1, \dots, x_N\}$ be an unlabeled evaluation pool and $E = \{e_1, \dots, e_H\}$ the submitted candidate entries. Entry e exposes response vector $a_e = (a_e(x_1), \dots, a_e(x_N))$ and has oracle utility vector u_e , binary per-example correctness in our experiments. An active policy π sequentially acquires B references. After step t , its transcript contains the queried indices, acquired references, and selected entries; using an oracle root map, we evaluate the corresponding root-level choices $\hat{r}_1, \dots, \hat{r}_t$.

Definition 1 (Candidate evidence frame). A frame $\Phi = (R, \rho, w)$ maps submitted entries to accountable candidate roots through $\rho : E \rightarrow R$ and assigns root mass w_r . A root is the unit that the evaluation claim intends to count once; it need not be observable or cryptographically authenticated. Two frames are response-compatible when they expose the same entry identifiers and response vectors.

Definition 2 (Registry refinement). A refinement of root r adds entries mapped to r without modifying existing entries. It is exact when an added response vector copies its parent and quality-equivalent when its oracle utility vector equals the parent’s coordinate-wise, even if their responses differ. Root-level refinement integrity requires a declared within-root rule under which such representation changes do not silently redefine the evaluation claim.

For example, a uniform per-entry mechanism induces the empirical measure $\mu_E = |E|^{-1} \sum_{e \in E} \delta_{a_e}$. Adding k entries changes one root’s represented mass without adding a root; candidate-consensus mechanisms can change analogously because consensus is computed over listed entries. A policy is *frame-blind* when it observes responses and acquired references but not authenticated ρ or w .

Table 1 compares strategic-replication bandits (Shin et al., 2022; 2025), active evaluation and selection (Lanctot et al., 2026; Liang et al., 2020; Nguyen et al., 2025; Okanovic et al., 2025; Kay et al., 2025; Durmazkeser et al., 2025; 2026), arena candidacy (Singh et al., 2025; Huang et al., 2026; Hays et al., 2026), clone weighting (George, 2010; Procaccia et al., 2025; Berriaud & Wattenhofer, 2026a), false-name work (Douceur, 2002; Burnat & Davidson, 2026), and lineage (Shang et al., 2026).

Table 1: Closest-work obligation boundary. These lines solve important pieces, but not the effect of an unaudited candidate unit on a complete adaptive selection transcript.

Line	Unit or evidence	Path?	Unmet obligation here
Replication bandits	Registered arms with known agent ownership	Yes	Shared labels, latent roots, terminal model choice
Sequential selection	Perturbed clones, pattern quotients, exact mergers, or supplied IDs	Yes	Shared labels, latent roots, exact-utility refinement, mediation
Arena candidacy	Provider/model clones	Final rank	Task-label acquisition transcript
Dilution/clone weights	Fixed neighborhood or metric	No	Root evidence and adaptive transcript
False-name/Sybil	Authority, resource, or evidence cluster	Not here	Active candidate-selection consequence
Model lineage	Ancestry evidence	No	Candidate mass and within-root rule

We do not claim the first exact-duplicate preprocessor, sequential replication or clone effect, registry effect, multiplicity attack, clone-proof weight, evidence-backed quotient, or lineage system. Our object is the shared-label sequential boundary: what predictions justify about response-distinct candidate units, how latent same-root multiplicity alters acquired reusable labels, and what root-level decision cost follows. Lineage is useful missing input, not a selection policy supplied here.

Natural refinement versus attack. Our natural anchor compares the original frozen roster with the exact-response quotient that removes its later-index version alias. It attributes neither common control outside the evaluation frame nor adversarial intent to the provider. It asks only whether a behaviorally redundant listed entry changes the active experiment. The adversarial tiers below are separate interventions that test whether an actor controlling one root can exploit the same entry-mass channel.

Offline threat tiers. The attacker controls one root and registers before acquisition. **T0** adds exact copies as a mechanism and defense control. **T1** learns on disjoint calibration references, then uses only holdout prompts, options, and parent outputs; quality is measured and approximately bounded, not assumed equivalent. **T2**, the main causal intervention, uses public or independently acquired references and changes only parent-wrong responses, exactly preserving correctness. Thus T2 fixes quality but assumes such reference access; T1 removes holdout-reference access and tests transfer. Across tiers, the attacker cannot modify existing entries, references, selector code, pool, budget, temperature, seeds, or tie-breaking, or adapt after acquisition begins.

Let $\ell(r)$ be root r 's full-pool loss and r^* the best root. We report ordered-path divergence, terminal regret $\ell(\hat{r}_B) - \ell(r^*)$, and the anytime quantity

$$\text{CReg}_B = \sum_{t=1}^B [\ell(\hat{r}_t) - \ell(r^*)].$$

Cumulative regret models repeated interim use; pairing it with terminal regret prevents a transient path effect from being misreported as a worse final choice.

3 FRAME IDENTIFIABILITY AND SEQUENTIAL AMPLIFICATION

We ask whether labels can compensate for a missing frame in any response-only policy, and whether a constant refinement can redirect a current selector. The first result establishes information necessity; the second gives concrete amplification.

Response-compatible worlds. Let $O(E) = \{(e, a_e) : e \in E\}$ be the public response view. Frames exposing the same $O(E)$ and label transcript are indistinguishable to a frame-blind policy: labels reveal useful behavior, not whether entries are independent roots. A fresh hidden-frame choice at every step makes this ambiguity accumulate.

For the construction below, let $V_{\Phi, B}^\pi$ be the expected information, in bits, acquired by policy π about B latent binary candidate coordinates, and let $V_{\Phi, B}^*$ be the optimum for a policy that knows Φ . Define $I_{\text{hi}} = 0.540852$ and $I_{\text{lo}} = 0.459148$ bits.

Theorem 1 (Tight cost of a missing evidence frame). *For every integer budget $B \geq 1$, there is one fixed public response registry, a pool with two binary queries per coordinate, and 2^B response-compatible frames such that every possibly randomized frame-blind policy has, on some fixed compatible frame,*

$$V_{\Phi, B}^* - V_{\Phi, B}^\pi \geq \frac{B}{2} (I_{\text{hi}} - I_{\text{lo}}) = 0.040852B \text{ bits.}$$

The bound is tight. The corresponding tight minimax excess coordinate-wise Hamming risk is $B/24$.

Proof intuition. One latent frame bit per coordinate selects which of two query polarities yields I_{hi} instead of I_{lo} . Under uniform compatible frames, an untouched bit is independent of prior observations, so a frame-blind policy chooses the informative polarity with probability at most one half. Revisiting cannot improve its minimax value. Averaging yields a fixed hard frame; independent randomization attains equality. Appendix B gives the construction and $1/24$ Hamming-risk calculation.

The construction grows exponentially with B , proving necessity rather than practical registry size. A constant change can nevertheless be amplified by a concrete selector.

Theorem 2 (Constant-registry full-path amplification). *For every odd $B \geq 3$ and finite Select-LLM temperature $\tau > 0$, there is a pool with five canonical response behaviors and $2B$ items such that adding four exact aliases produces nine submitted entries, redirects all B acquisitions to a disjoint query block, and changes cumulative regret from zero to at least*

$$\frac{B^2 - 1}{4B} = \Omega(B),$$

while the terminal regret is $(B - 1)/(2B)$. The two best canonical behaviors and all existing responses are unchanged.

Proof intuition. The clean registry always prefers a good query block. Four aliases move another behavior’s mass above one half along an alternating-label path, making every bad query preferable and reinforcing the condition after each selection. Counting odd steps gives both regret bounds; Appendix B verifies them for every finite τ .

This is mechanism-specific, not universal harm, and exact hashing blocks the literal construction. Section 4 therefore tests response-distinct, quality-equivalent variants.

Proposition 1 (Selector-agnostic root factorization). *Suppose ρ is authenticated, each root has fixed mass w_r , and at every step a selector’s state, action kernel, and decision depend on entries only through root summaries S_r . If S_r is invariant to an allowed within-root refinement, then the distribution of the complete query transcript and root decisions is invariant to that refinement.*

Transcript induction proves the claim (Appendix B). It covers uncertainty, diversity, and mutual-information rules only when they actually factor through authenticated, refinement-invariant root summaries; a method label alone gives no protection. Exact quotienting handles unique response patterns, not response-distinct variants. Predictions also cannot authenticate ρ , so deployment still requires a declared canonical entry or fixed within-root mixture.

4 LOCKED INTERVENTION AND EVALUATION PROTOCOL

Primary selector and response matrices. Select-LLM uses one similarity s for both reference–output posterior evidence and output–output acquisition (Durmazkeser et al., 2026). Our independent *exact-answer primary specialization* sets $s(a, b) = \mathbf{1}[a = b]$ in both places. With a uniform prior over submitted entries, its posterior and acquisition score are

$$p_t(e) \propto \exp\left(\tau^{-1} \sum_{s < t} s(r_{i_s}, a_e(x_{i_s}))\right), \quad A_t(x) = \sum_y \left(\sum_{e: a_e(x)=y} p_t(e) \right)^2.$$

At each step it queries the unlabelled item minimizing A_t , observes its reference, and selects a root with maximum accumulated exact-match evidence. This source-fidelity claim applies to the natural anchor and primary exact-answer T2, not to later F1/BLEU or confidence-view transfer audits. To make the primary implementation independent of submitted-list order, V2 canonicalizes entries by full-response SHA-256, root ID, and label hash; a locked cross covers all nine exact query/root tie policies. Thus only the registry differs between paired clean and refined runs.

We use raw responses and recorded reference-based scores for 31 models from HELM-Lite v1.0.0 (Liang et al., 2023) on MedQA (Jin et al., 2021), GSM8K (Cobbe et al., 2021), and OpenBookQA (Mihaylov et al., 2018). Table 2 records the complete primary mapping.

Table 2: Frozen primary LLM protocol. Each final condition uses 500 paired pool draws.

Task	N	Pool	Budgets	τ	Development-locked parent
MedQA	1000	500	10/30/50	3	GPT-4-1106-preview
GSM8K	1000	500	10/20/30	5	GPT-4-0613
OpenBookQA	500	400	10/20/30	5	text-davinci-002

Natural exact-alias anchor. Before base-versus-refined outcomes, we select the unique pair whose full response and utility rows match in every task: frozen indices 9 and 10. The stable-order 30-entry base removes index 10; the original roster restores it and maps it to root 9. Primary settings and 500 paired pools follow Table 2; five-temperature and three-budget slices give 21 cells. Pair discovery used row equality but not selector outcomes, making this a result-blind internal lock rather than an external prospective preregistration. A locked replay separates ordered displacement from unique-set substitution; Appendix D gives its descriptive rules.

Quality-equivalent refinement. Our realism confirmation uses an answer-space-native T2-A constructor. For each of four aliases, a frozen hash orders positions where parent e is wrong and selects $d = \max\{1, \text{round}(\eta N)\}$. On multiple-choice tasks it emits another public option that is

neither e 's answer nor the reference; on GSM8K it emits a deterministic nonzero numeric offset from the reference. Thus every alias has a distinct response hash while satisfying $u_{e'} = u_e$ coordinate-wise. Offline construction reads e 's outputs, public IDs/options, and references—no peer response, pool seed, or trajectory—before pool draws. A locked replay then serves prompt-only wrappers with no runtime ID, reference, peer response, pool seed, or selector state.

Targets were locked earlier by a matrix-based T2-M stress constructor over every eligible parent on development pools 3500–3999. Its target distribution, robustness, and defenses use seeds 50000–50499. Without retargeting, T2-A uses fresh seeds 91000–91499. These are new pool realizations of the same public benchmarks, not disjoint tasks or clinical cohorts.

Prospective reference-free T1. Before content or outcomes, we froze eight untouched HELM-Lite scenarios: NarrativeQA, two NaturalQA modes, and five WMT14 pairs. Calibration alone trains a TF-IDF ridge error score and fixes four abstention wrappers; the builder reads no holdout reference, utility, peer output, pool, or selector state. Bounded utilities make every alias non-improving and at most one point worse. These tests intentionally use exact-response grouping for acquisition but task-native F1/BLEU for posterior evidence and regret, so they audit a *decoupled-view selector*, not source-faithful Select-LLM. On 1,000 paired pools per scenario, corrected gates require terminal and active-minus-fixed harm. Every null is retained (Appendix A).

Learned transfer and source-fidelity confirmation. Step 92's learned AG News raw-text adapters use confidence acquisition but hard-label posterior evidence, so this is a *decoupled-view* audit (Appendix H). Steps 93–94 instead use the same literal $s(a, b) = 1[a = b]$ for acquisition and evidence. Four learned gates emit their parent label or a unique out-of-label abstention token. Step 93 freezes 504 Banking77 (Casanueva et al., 2020) conditions before 1,000 test pools; Step 94 freezes three unused tasks and development-selects one roster before 2,000 20 Newsgroups pools. Both require .5 terminal point, positive inference, $\geq 50\%$ path change, at most one-point loss, and exact fixed-query equality; all failures are retained (Appendices I–J).

Targeted-intervention pairing, robustness, and inference. Each seed samples a pool without replacement; its clean and refined runs share the pool, references, budget, temperature, and tie-breaking. Temperatures were selected without references on disjoint seeds 40000–40499 by maximizing mean proxy identification probability over steps $1:B$ on the frozen grid $\{0.1, 0.2, \dots, 1, 2, 3, 4, 5\}$, with smaller- τ tie-breaking. The robustness audit crosses five temperatures $\{0.1, 0.5, 1, 3, 5\}$, four alias counts $\{1, 2, 4, 8\}$, four distances $\{0.1\%, 1\%, 5\%, 10\%\}$, and three budgets per task. It retains all 660 eligible cells; an infeasible distance is marked ineligible without changing the parent.

The primary endpoint is paired change in cumulative fixed-universe deployed-output regret (Section 2); terminal regret, ordered-path divergence, changed positions, query-set Jaccard, and harmful/improved/tied pool fractions are secondary. We use 10,000-repetition paired bootstrap intervals and paired two-sided sign-flip tests, with Benjamini–Hochberg correction within each prespecified family. Uniform random querying tests whether active acquisition adds clean value. A second control fixes the same query matrix in both registries and decomposes static root selection from sequential excess harm. Sixteen frozen permutations of each clean/refined registry require bitwise equality of query, root, regret, and acquisition-value paths after canonicalization. All negative cells remain in the reported grid.

Matched defense audit. The exact quotient groups identical full response vectors. For radius ρ , the behavioral cover connects entries whose full-vector disagreement is at most ρ , takes connected components, and retains the smallest-index representative. We set ρ to the alias's realized distance—never a larger unmatched radius—and jointly report residual attack effect, clean cumulative and terminal cost, models removed beyond exact quotienting, natural same-provider merges, and maximum within-component quality span. Authenticated lineage that assigns fixed parent-level mass is an oracle-information comparator, not a proposed response-only defense.

We also preregister three general-metric clone-robust graph weights from Berriaud & Wattenhofer (2026b): class-uniform, maximal-clique equal-allocation (MCCA), and maximal-clique participation (MCCP). Under normalized response Hamming distance, exact integration over a uniform radius density and seven frozen α values supplies the prior for both acquisition and deployment, yielding 63 clean and 105 attack cells. The original “suppressed” gate requires at most 10% residual regret

and path change; “clean-preserving” requires $|\Delta \text{CReg}| \leq 0.1$ and both intervals to contain zero. We retain those integrity gates, but also report a post-hoc descriptive fixed-configuration audit of outcome residual and clean cumulative/terminal cost; path identity is not treated as necessary for equal utility.

Official cross-selector audits. We retain an official CODA path-sensitivity audit on five GLUE tasks (Appendix E). Separately, before outcomes, we froze all six official LLM Selector datasets (Durmazkeser et al., 2025) and each source-order-first parent. Four entries copy its complete strong-judge row but hash-cycle 20% of only its own weak-judge tensor; no other candidate, strong label, pool, or trajectory is read. We reproduce the official ten-judge entropy rule for 30 labels and 1,000 paired pools per dataset. Promotion requires exact utility and fixed-query equality, material terminal harm with corrected inference, and two passing datasets. This is a score-matrix decision-view intervention, not a raw-response endpoint (Appendix F).

5 RESULTS

5.1 A NATURAL EXACT ALIAS REWRITES THE ACTIVE EXPERIMENT

The two frozen version entries are the unique cross-task exact response-and-utility pair: zero mismatches over 2,500 items and equal task qualities .548/.721/.844. Restoring the later entry to its 30-entry quotient adds no behavior or quality vector and is not attributed to adversarial intent, yet changes at least one query in all 500/500 runs per task. Mean changed positions and query-set Jaccards are 40.744/.844, 21.752/.911, and 16.010/.940 (Table 3, Panel A). With candidate-invariant random queries, query matrices, root histories, and regret remain exactly equal. The active effect follows from raising one behavior’s entry mass from 1/30 to 2/31.

Table 3: Natural and targeted refinements on 500 paired pools per task. Panel A restores one naturally occurring exact version alias to its 30-entry quotient. Panel B reports the canonical V2 replay of fresh-pool, 5%-distance T2-A effects constructed without peer responses. Every run changes path; CReg is summed through *B* and positive is worse.

A. Natural exact alias: occurrence and representation non-invariance					
Task	Changed positions	Query Jaccard	Δ CReg [95% CI]		Unique repl.
MedQA	40.744/50	.8437	−0.5002 [−0.6152, −0.3883]		4.358 (8.72%)
GSM8K	21.752/30	.9111	+0.0593 [0.0275, 0.0921]		1.420 (4.73%)
OpenBookQA	16.010/30	.9395	−0.0276 [−0.0634, 0.0083]		0.960 (3.20%)
B. Target-locked, peer-response-free T2-A: targeted exploitability					
Task	B	Δ CReg [95% CI]		Δ terminal pp [95% CI]	Harmed pools
MedQA	50	+3.2518	[3.0214, 3.4842]	+2.906 [2.316, 3.543]	92.0%
GSM8K	30	+3.7811	[3.6177, 3.9433]	+10.559 [9.908, 11.234]	99.2%
OpenBookQA	30	+0.8459	[0.7650, 0.9260]	+0.006 [−0.016, 0.031]	78.8%

Ordered displacement overstates acquired-evidence replacement. Query sets change in 94.0/87.4/66.0% of runs, but only 4.358/1.420/.960 unique labels on average (8.72/4.73/3.20% of budget); only MedQA exceeds our disclosed descriptive 5% heuristic. Natural CReg improves by .5002, worsens by .0593, and is inconclusive, respectively, while all primary terminal intervals contain zero. All 21 cells remain path-material and 14 CReg effects survive correction (five harms, nine improvements). We therefore claim cross-task transcript instability and task-local label substitution, not broad annotation-cost or reliable natural harm.

5.2 TARGETED RESPONSE-DISTINCT REFINEMENTS CAUSE FRESH-POOL HARM

The intervention changes representation, not recorded quality. Without retargeting, every T2-A alias has a distinct non-parent response hash yet exactly preserves its parent’s correctness on every example. Original entries, references, and the best original root are unchanged. The constructor reads zero peer-response rows. Prompt-only replay is exact for all 12 alias rows and 1,500 runs (2.3–5.1 KB manifests; zero overrides after a one-character mutation). Nevertheless, all 500 fresh paired trajectories change on every task. Fixing one random-query matrix for both registries makes query histories and regret exactly equal, so the entire paired penalty is sequential excess rather than pool resampling or registry-only deployment choice.

T2-A increases cumulative deployed-output regret on all three fresh confirmation sets (Table 3, Panel B); all intervals exclude zero and BH-adjusted sign-flip q -values equal 9.999×10^{-5} . The final selected root changes on 74.4% of MedQA pools and 80.6% of GSM8K pools, yielding terminal losses of 2.91 and 10.56 percentage points. OpenBookQA’s terminal interval contains zero despite +0.846 cumulative regret, supporting an anytime rather than final-model failure.

The effect is not a candidate-order or favorable-tie artifact. Across 16 frozen permutations of both registries, all 48,000 scenario–pool replays have zero query, selected-root, regret, acquisition-value, or canonical-digest mismatch. All 27 cells in the 3×3 exact query/root tie grid change every path and have positive cumulative-regret intervals: mean effects range from 2.089–3.252 on MedQA, 3.781–3.884 on GSM8K, and .415–1.340 on OpenBookQA. Every query policy has positive terminal harm under all three root policies on MedQA and GSM8K; OpenBookQA remains a cumulative-only result. Clean V2 active selection beats random querying in cumulative regret on GSM8K and OpenBookQA; on MedQA its cumulative difference is inconclusive, but exact-identification AUC improves by .129 [.090,.167]. Thus the attack is measured on a nonvacuous active selector without relying on insertion order.

Reference-free terminal harm transfers prospectively to three of eight untouched scenarios. After calibration locked every configuration, four distinct aliases per scenario were built from holdout prompts and parent outputs alone and stayed non-improving within one utility point. Every fixed-query effect is exactly zero. NarrativeQA terminal regret rises by 3.021 points [2.766,3.282], and WMT14 fr–en and ru–en rise by .139 [.102,.177] and .250 [.154,.350] BLEU-4 points; both active and active-minus-fixed eight-task BH $q \leq .00027$. Their paths change in 100.0/75.1/97.4% and final roots in 45.6/10.6/46.1% of 1,000 paired pools. The other five frozen scenarios are retained: NaturalQA closed/open and WMT de–en improve, while cs–en and hi–en intervals cross zero. The preregistered decision is therefore 3/8 PROMOTE_REFERENCE_FREE_TERMINAL_BRIDGE, not a universal transfer claim (Appendix A).

Learned raw-input transfer is positive under a decoupled view; same- s cumulative harm replicates but terminal gates fail. On sealed AG News, 50,817-parameter decoupled-view adapters preserve every parent answer, change every path, and raise terminal regret by .547 [.502,.593] points (Appendix H). Under the stricter single s , Banking77 aliases lose .098–.358 points, change 99.8% of paths, and raise CReg by .0409 [.0129,.0689], but terminal harm is +.032 [−.055,.122]. A three-task-sealed protocol then development-selects 20 Newsgroups: integrated 205,245-parameter raw-text variants lose .252–.491 points, change every path and 48.2% of terminal roots, and raise CReg by .0822 [.0385,.1266] ($p = .0002$); terminal harm is +.120 [−.044,+.290] ($p = .081$). Both single- s audits have matching active-minus-fixed effects and exact fixed-query equality. Thus learned anytime harm replicates, but neither passes the .5-point terminal gate (Appendices I–J).

Selection is reproducible but the effect is bidirectional. In the T2-M stress audit, parents chosen only on development pools rank first by harm on final pools; development–final target-effect Spearman correlations are 0.991, 0.995, and 0.999. Yet its complete target audit rejects universal harm. Positive versus negative point effects are 15/16 on MedQA, 13/18 on GSM8K, and 12/18 on OpenBookQA; the corresponding medians are −0.006, −0.046, and −0.176. Hence the result is a development-selectable directional intervention, not an average claim about arbitrary refinement.

The targeted effect is robust, not monotone. For T2-M at 5% distance and four aliases, all 15/15 task–temperature cells and 9/9 task–budget cells have positive BH-corrected intervals. Across the complete 660-cell grid, MedQA has 240/240 positive cells and GSM8K 180/180 eligible positive cells. OpenBookQA has 171 positive and 69 negative cells (167 and 60, respectively, BH-significant): at its primary budget and temperature, four 5% aliases give +1.460 cumulative regret whereas eight give −0.438. T2-A at 1% is likewise positive on MedQA/GSM8K but improves OpenBookQA by .545. Alias count and distance are therefore not scalar corruption strengths; registry mass can steer the trajectory in either direction.

5.3 PREDICTION-ONLY DEFENSES MITIGATE OUTCOMES BUT DO NOT TRANSFER CLEANLY

Exact quotienting removes the natural pair and T0 copies but retains response-distinct aliases. A matched 5% cover suppresses T2-M at clean CReg cost 0.000/0.000/ − 0.169 on MedQA/GSM8K/OpenBookQA. At 10%, GSM8K is ineligible; OpenBookQA merges six natural entries spanning 5.2 accuracy points and pays 3.975 terminal points, while MedQA pays only cumulative cost. Authenticated lineage has zero clean change but uses external information (Appendix L).

Soft weights reduce absolute CReg in 74/105 cells, but none is non-worsening across all attacks. Best-mean MCCA ($\alpha = .132$) leaves .584 residual, worsens GSM8K by 15.8%, and changes clean CReg by −0.180/ +0.386/ +1.273; 45/63 terminal intervals contain zero. Every attacked transcript differs: a task-dependent frontier, not universal impossibility.

5.4 CROSS-SELECTOR EFFECTS INCLUDE TERMINAL HARM

The official LLM Selector audit changes every path, preserves strong-judge utility, and has zero fixed-query effects. AlpacaEval/Bingo pass: terminal regret rises by .321 [.194, .453]/.498 [.374, .623] points (six-task BH $q = 3.0 \times 10^{-5}$), with 23.2/32.4% final-root change. Arena-Hard improves, Flickr30k misses correction, and MEDIQA/MT-Bench are terminal-null. These are weak-view aliases (Appendix F); CODA has mixed cumulative and negligible terminal effects (Appendix E). Thus harm crosses a second family without becoming universal.

Negative findings bound the conclusion. Natural, OpenBookQA, and CODA cells improve. Only 8/420 untargeted additions are materially harmful versus 27 beneficial, and a fragility score fails five of six gates (Appendix K). We claim targeted, not average, harm.

6 RELATED WORK

Closest conceptually, strategic-replication bandits let known agents register duplicate arms and motivate agent-first defenses (Shin et al., 2022; 2025). They observe arm-local rewards under supplied ownership; here one label scores every entry, roots are latent, and the object includes terminal root choice. Active evaluation tests perturbed, separately ranked clones; UNREAL and PUTSC quotient redundant committees or candidates (Lanctot et al., 2026; Nguyen et al., 2025; Liang et al., 2020). We claim neither the first sequential replication nor clone effect. Our remaining object is shared-label exact-utility refinement with fixed-query mediation to root regret (Table 1); correlation-aware selection fixes an ensemble (Turkmen et al., 2026).

Static registry attacks change ranks, models, matches, or descriptions, not future labels (Hays et al., 2026; Suri et al., 2025; Singh et al., 2025; Huang et al., 2026). Dilution priors, clone weights, quotients, and lineage constrain static influence or identify roots (George, 2010; Berriaud & Wattenhofer, 2026b; Procaccia et al., 2025; Burnat & Davidson, 2026; Shang et al., 2026); ours changes which reusable labels are acquired, which post-hoc quotienting cannot recover.

7 LIMITATIONS, IMPLICATIONS, AND CONCLUSION

The natural anchor is one pair; T2-A uses public references; T1 passes 3/8, LLM Selector 2/6, and untargeted additions are mostly benign. AG News is decoupled-view; learned single- s Banking77/20 Newsgroups have positive CReg but null terminal harm, and neither is a full checkpoint or registry admission. We establish targeted evidence-frame failures—not production compromise, clinical harm, universal terminal damage, intent, or the first sequential clone effect. Registries should bind roots and root mass (Proposition 1), or report evidence-frame and multiplicity sensitivity.

AI USE STATEMENT

Generative AI tools assisted topic and gap discovery, conceptual framing and hypothesis refinement, literature discovery and organization, methodology and experiment-design critique, code drafting and experiment orchestration, formulation and checking of formal statements and proof drafts, result

analysis and interpretation, adversarial manuscript review, artifact documentation, and language editing. AI outputs were not treated as sources of evidence or as autonomous authorship. The authors manually checked citations against primary sources, executed and validated the code and locked outputs, inspected the proof obligations, reviewed all AI-assisted text, and take responsibility for the complete submission.

ETHICS STATEMENT

The work audits public benchmark responses and does not use clinical records or interact with deployed patients. We disclose attack prerequisites and negative results to avoid overstating private-label or clinical risk. The intended use is to improve the integrity of evaluation registries and active selection protocols.

REPRODUCIBILITY STATEMENT

The anonymous artifact records the natural-alias lock, preregistered targets, paired seeds, complete robustness and defense grids, official-selector provenance, and machine-readable outputs. Its manifest binds Steps 80/87/88, configuration-wrapper Steps 89–90, learned decoupled-view Step 92, and source-faithful Steps 93–94, including their learned weights, sealed negative results, complete development ledgers, and independent adjudications. Offline mode verifies every bundled byte and anonymity. The top-level full mode verifies inherited manifest and receipt bindings, rebuilds the paper, and independently reconstructs the current learned single-*s* outcomes. It does not transitively rerun earlier-stage validators; those remain separately invocable for Steps 80/87/88/90/92. Public source packages, base weights, and third-party repositories are not embedded.

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A PROSPECTIVE REFERENCE-FREE TERMINAL AUDIT

Outcome-blind task freeze. An initial preregistration named 18 fresh MMLU subjects, but all 31 model-prefix requests for its first frozen subject returned HTTP 404 before any response or outcome was observed; HELM-Lite v1.0.0 contains only the five MMLU subjects already used elsewhere in the project. We retained that structural failure rather than replacing a result. A metadata-only enumeration then identified eight previously untouched scenarios with the complete 31-model roster. Before downloading their content or outcomes, a separate protocol froze NarrativeQA, NaturalQA closed/open, and WMT14 cs/de/fr/hi/ru-en. These exhaust the untouched, structurally complete non-LegalBench HELM-Lite scenarios left after excluding prior tasks; no frozen scenario could be dropped or replaced.

Each task was split by a manifest-keyed SHA-256 ordering: NarrativeQA contributes 235 calibration and 235 sealed-holdout items, and every other scenario contributes 500 and 500. The first metric present for every model-item pair under the frozen order F1, quasi-exact match, exact match, BLEU-4, BLEU was selected without attack outcomes; this yields F1 for the three QA scenarios and BLEU-4 for translation. Utilities lie in $[0, 1]$.

Attack freeze and gate. Five-fold out-of-fold TF-IDF bigram ridge prediction estimates the selected parent’s continuous error from prompt and parent output. A prediction-only exact-clone screen retains five parents. Calibration then selects among three support-overlap rules, distances .5/1%, budgets 10/20/40, and clean-selected temperatures 1/3/5 by maximizing active-minus-fixed

terminal harm over 200 paired pools. Only after that choice does the builder open the selected parent’s blind holdout prompts and outputs. It changes the top predicted-error coordinates to four alias-specific opaque abstention strings; no holdout reference, utility, peer output, pool, trajectory, or selector state is read. Unchanged utilities are inherited and changed utilities are zero, so every alias is coordinate-wise non-improving and loses at most its declared distance. The tokens are subsequently checked against all sealed references and have zero collision.

Stage B runs exactly 1,000 paired pools (seeds 88400–89399) per frozen scenario. A scenario passes only if (i) all aliases remain within one percentage point, (ii) both active terminal harm and active-minus-fixed terminal harm have paired-bootstrap 95% lower bounds above zero, and (iii) their separately eight-task BH-adjusted one-sided sign-flip q -values are below .05. Fixed-query terminal effects are reported separately; any nonzero value forbids exact mediation. At least two passing scenarios were required for promotion.

Table 4: Prospective reference-free terminal audit on eight previously untouched HELM-Lite scenarios. Values are utility percentage points; q^* is the larger of the two eight-task BH-adjusted one-sided sign-flip q -values for active terminal harm and active-minus-fixed terminal harm. Clean and refined fixed-query effects are exactly zero in every row. All rows were frozen and retained.

Scenario (metric)	d	B	Quality gap	Δ terminal [95% CI]	q^*	Path/root change	Gate
NarrativeQA (F1)	1%	10	−.851	+3.021 [2.766, 3.282]	.00027	100.0/45.6%	Pass
NaturalQA closed (F1)	.5%	10	.000	−1.213 [−1.443, −.985]	1.000	97.6/55.5%	No
NaturalQA open (F1)	1%	10	−.842	−.317 [−.552, −.075]	1.000	99.6/35.6%	No
WMT14 cs-en (BLEU-4)	1%	10	−.109	+ .080 [−.033, .192]	.161	99.1/39.5%	No
WMT14 de-en (BLEU-4)	1%	10	−.275	−.087 [−.169, −.002]	1.000	94.2/35.9%	No
WMT14 fr-en (BLEU-4)	1%	40	−.064	+ .139 [+ .102, .177]	.00027	75.1/10.6%	Pass
WMT14 hi-en (BLEU-4)	1%	20	.000	+ .001 [−.033, .035]	.757	100.0/15.4%	No
WMT14 ru-en (BLEU-4)	1%	10	.000	+ .250 [+ .154, .350]	.00027	97.4/46.1%	Pass

Three scenarios pass: NarrativeQA, WMT14 fr-en, and WMT14 ru-en. Their worst alias quality gaps are −.851, −.064, and .000 percentage points, respectively; thus the Russian pair is exact-utility-equivalent on the sealed holdout. All eight fixed-query root histories and regret arrays are byte-identical, making the active and active-minus-fixed point effects equal. The five nonpassing scenarios are not hidden: two confidence intervals cross zero, while NaturalQA closed/open and WMT14 de-en significantly improve the selected terminal utility. The machine decision is PROMOTE_REFERENCE_FREE_TERMINAL_BRIDGE; it supports a selectable, reference-free attack on new task families, not monotone transfer.

The Stage-A attack was frozen before deserializing the sealed outcomes. The Stage-B lock binds the preregistration, attack rows, sealed package, runner, independent validator, and six runtime dependencies. A separate implementation reconstructs canonicalization, active and fixed acquisition loops, every raw path/regret array, 10,000-repetition intervals and sign-flip tests, BH adjustment, quality/integrity gates, and the final 3/8 decision. All eight task checks, all four lock checks, and the decision check pass under PASS_STEP87B_INDEPENDENT_RECONSTRUCTION. The immutable results and raw-array SHA-256 values begin FC953C87 and 26422130; the independent-validation hash begins 1EF6C0F7.

B FULL PROOFS AND FORMAL VERIFICATION

B.1 PRODUCT-FRAME CONSTRUCTION

Fix budget B and let each canonical behavior be a vector $z \in \{A, B\}^B$. The public registry contains 2^B exact submitted copies of every z , for 4^B submitted IDs in total. A latent frame is indexed by $\omega \in \{A, B\}^B$. Define

$$m_\omega(z) = |\{t : z_t = \omega_t\}|.$$

Frame Φ_ω partitions the 2^B public copies of z into $r_\omega(z) = 2^{m_\omega(z)}$ accountable roots, each containing $2^{B-m_\omega(z)}$ aliases. These are integer partitions of the same public leaves. Every frame contains

$$\sum_z 2^{m_\omega(z)} = 3^B$$

roots. A uniform root prior therefore induces

$$\Pr_{\omega}(F = z) = \frac{2^{m_{\omega}(z)}}{3^B} = \prod_{t=1}^B \begin{cases} 2/3, & z_t = \omega_t, \\ 1/3, & z_t \neq \omega_t. \end{cases}$$

The candidate coordinates are independent, with $\Pr(F_t = \omega_t) = 2/3$, although the submitted IDs and responses are identical in all 2^B frames.

For each coordinate t , the fixed pool contains two binary queries $q_{t,+}$ and $q_{t,-}$. With $c = 1/4$, their channels are

$$\begin{aligned} q_{t,+} : \quad & \Pr(Y = 1 \mid F_t = A) = 1, & \Pr(Y = 1 \mid F_t = B) = c, \\ q_{t,-} : \quad & \Pr(Y = 1 \mid F_t = A) = 1 - c, & \Pr(Y = 1 \mid F_t = B) = 0. \end{aligned}$$

Outcomes are conditionally independent across coordinates given F . If $\Pr(F_t = A) = 2/3$, direct substitution in $I(F_t; Y) = H(Y) - H(Y \mid F_t)$ gives

$$I(F_t; Y_{t,+}) = 0.540852 =: I_{\text{hi}}, \quad I(F_t; Y_{t,-}) = 0.459148 =: I_{\text{lo}};$$

the values reverse when the prior is $1/3$.

B.2 PROOF OF THEOREM 1

Place a uniform distribution over the 2^B frames. Until coordinate t is queried, ω_t is independent of every observation from the other coordinates. Thus any frame-blind policy chooses the correct polarity on a fresh coordinate with average probability at most $1/2$, and obtains at most

$$\frac{1}{2}(I_{\text{hi}} + I_{\text{lo}}) = 0.5$$

bits on its first query to that coordinate.

It remains to show that adaptivity and repeated queries do not improve this average bound. The prior entropy of one coordinate is $h(2/3) = 0.918296$ bits. Even after the lower-information first query, its remaining entropy is at most

$$h(2/3) - I_{\text{lo}} = 0.459148 < 0.5.$$

Consequently, even a hypothetical follow-up that revealed all remaining uncertainty is worse than the average value of querying a fresh coordinate. A frame-aware policy also moves to a fresh coordinate because $h(2/3) - I_{\text{hi}} = 0.377444 < I_{\text{hi}}$. An exchange argument therefore transforms either optimal policy into one that queries each coordinate exactly once without reducing value.

The frame-aware value is BI_{hi} , whereas the average frame-blind value is at most $B/2$. Hence the average regret over frames is at least

$$B(I_{\text{hi}} - 1/2) = \frac{B}{2}(I_{\text{hi}} - I_{\text{lo}}),$$

so at least one fixed frame attains the claimed lower bound. A policy that visits each coordinate once and independently randomizes its polarity obtains exactly $(I_{\text{hi}} + I_{\text{lo}})/2$ on every frame, proving tightness.

For Hamming loss, the posterior Bayes error is $1/12$ after the correct-polarity channel and $1/6$ after the wrong-polarity channel. Fair randomization gives $1/8$. An untouched coordinate has error $1/3$, so a fresh query reduces error by $5/24$; all error remaining after a first query is at most $1/6 < 5/24$. The same exchange argument applies, and the minimax excess is

$$B(1/8 - 1/12) = B/24.$$

B.3 PROOF OF THEOREM 2

Let $B \geq 3$ be odd. The five canonical candidates are A, B, C, D, E , and the pool has B good and B bad binary items. On every good item their responses are $(0, 0, 1, 1, 1)$ and the reference is 0. On bad item t their responses are $(0, 0, 1, 0, 0)$, with reference 1 for odd t and 0 for even t . The refinement adds four exact copies of C .

For Select-LLM’s exact-match acquisition, write

$$S(x) = x^2 + (1 - x)^2,$$

where x is posterior mass predicting one. After s clean good labels, A and B have likelihood weight $a = e^{s/\tau}$ and C, D, E have weight one. With $d = 2a + 3$, an unqueried good item has one-mass $3/d$, while a bad item has one-mass $1/d$. Since

$$S(1/d) - S(3/d) = \frac{4(d-4)}{d^2} > 0,$$

every remaining good item has strictly lower acquisition score. The clean selector exhausts the good block, and A, B remain selected.

In the refined registry, behavior C has five of nine entries. Along the bad path, its aggregate posterior mass is $x = 5/9$ after every even number of bad labels and

$$x = \frac{5e^{1/\tau}}{5e^{1/\tau} + 4}$$

after every odd number, so $1/2 < x < 1$ for every finite $\tau > 0$. A bad item has one-mass x . A good item groups C with D, E and has one-mass $(1+x)/2$. The bad item is strictly closer to one half because $x - 1/2 < x/2$ whenever $x < 1$, and hence has lower S . Induction shows that the refined selector exhausts the disjoint bad block.

Across the complete $2B$ -item pool, A and B outperform C by

$$\Delta_B = \frac{B-1}{2B}.$$

After each odd bad label, C has one more observed correct response than every non- C candidate, so the selected behavior must be C . There are $(B+1)/2$ such steps. Even granting favorable tie-breaking on every even step, cumulative regret is at least

$$\frac{B+1}{2} \Delta_B = \frac{B^2-1}{4B}.$$

The final step is odd, giving terminal regret Δ_B .

B.4 PROOF OF PROPOSITION 1

Consider any allowed refinement and couple the clean and refined runs using the same policy randomization and reference randomness. Initially, fixed root masses and refinement-invariant summaries give identical root-level states. Assume the transcripts agree through step $t-1$. The selector’s state is then the same by factorization, hence so is its conditional action kernel. Under the coupling it selects the same query and observes the same reference; refinement invariance produces the same updated root summaries and root decision. Induction gives identical complete transcripts and decisions almost surely, hence equal distributions. The argument applies to any acquisition objective—including uncertainty, diversity, or mutual information—only if its implemented state and action kernel satisfy the stated factorization.

B.5 EXECUTABLE VERIFICATION

The standard-library checker verifies the channel information values and Bayes errors; every integer root partition for $B = 1, \dots, 6$; all deterministic polarity vectors against all frames for $B = 1, \dots, 8$; and the amplification instance for $B \in \{3, 5, 11, 51\}$ and $\tau \in \{.1, .5, 1, 3, 10\}$. All assertions pass. At $B = 51$, every tested temperature gives clean cumulative regret zero, refined cumulative regret 12.745098, terminal regret .490196, and 51/51 acquisitions redirected to the disjoint block.

C COMPLETE LLM PROTOCOL AND RESULTS

T2-M target selection used development pools 3500–3999 and was complete over every eligible parent; primary outcomes used paired pools 50000–50499 only after the parent lock. The complete robustness

cross is temperature $\{.1, .5, 1, 3, 5\}$, alias count $\{1, 2, 4, 8\}$, response distance $\{.1\%, 1\%, 5\%, 10\%\}$, and the three task-specific budgets in Table 2. This yields 240 MedQA, 180 GSM8K, and 240 OpenBookQA cells. GSM8K’s locked parent is correct on too many items to construct quality-identical 10% aliases, so all 60 such cells are marked ineligible rather than retargeted.

Table 5: Complete T2-M grid; every eligible cell is retained. Positive and negative indicate the sign of mean paired cumulative-regret change.

Task	Eligible	Positive	Negative	BH-significant $+/-$
MedQA	240	240	0	240/0
GSM8K	180	180	0	180/0
OpenBookQA	240	171	69	167/60

At 5% distance, four aliases, and the primary budget, all five temperatures are positive in all three tasks. At each locked temperature, all three budgets are likewise positive, for 24/24 task-setting cells. OpenBookQA supplies the important non-monotone boundary: at its primary setting, four 5% aliases give $+1.460$ cumulative regret but eight give $-.438$. We report all cells because alias count and distance change a directional prior; they are not scalar corruption strengths.

Every final pool is sampled without replacement and shared by its clean/refined pair. The paired bootstrap and sign-flip tests use 10,000 repetitions; Benjamini–Hochberg correction is applied within each prespecified family. Target development/final mean-effect rank correlations are $.991/.995/.999$ for MedQA/GSM8K/OpenBookQA. The raw artifact stores the 500 paired cumulative and terminal arrays, query paths, selected roots, hashes, and eligibility reason for every cell.

Peer-response-free realism confirmation. After the T2-M targets were frozen, a separately hash-locked audit rebuilt four 5%-distance aliases from only the parent’s outputs, public IDs/options, and references. It used no peer response or response-frequency statistic and generated pools only after construction. Under the historical backend, fresh seeds 91000–91499 gave mean paired $\Delta CReg$ of 3.2657 [3.0341, 3.4997], 3.7851 [3.6247, 3.9492], and .9552 [.8742, 1.0342] on MedQA, GSM8K, and OpenBookQA. Fixed-query paired effects were exactly zero. A sequential-reference variant stopped after 258/1000, 697/1000, and 136/500 inspected labels; it harmed the first two tasks but improved OpenBookQA by .6805. The preregistration, raw 500-pair arrays, and isolated-constructor replay are stored in the Step 54 artifacts; the validator reports `PASS_STEP54_MATRIX_FREE_REALISM_AUDIT`.

Canonical V2 implementation-stability transfer. An implementation audit found that evidence from a separately tested legacy matrix-aware wrapper could not certify the paper’s matrix-free headline condition. Before computing V2 outcomes for that exact condition, we therefore hash-locked its unchanged aliases, data, pools, targets, budgets, and temperatures, plus canonical entry ordering and an exact 3×3 tie-policy cross. The V2 replay gives 3.2518 [3.0214, 3.4842], 3.7811 [3.6177, 3.9433], and .8459 [.7650, .9260]. Fixed-query effects remain exactly zero, so sequential-excess effects equal the full paired penalties, with independently recomputed positive lower bounds of 3.0231, 3.6209, and .7649. All 16 permutations of both registries produce zero bitwise mismatch over 48,000 scenario–pool paths. Every one of the 27 tie cells changes all paths and has a positive cumulative interval; MedQA/GSM8K terminal means are positive under every tie policy, while OpenBookQA is not promoted as terminal harm. A runner-independent reconstruction replays aliases, acquisition paths, 10,000-repetition inference, BH correction, and adjudication in 2,882 checks, reporting `PASS_STEP80_INDEPENDENT_VALIDATION` and `STRONG_TRANSFER`. Main-text values use this V2 backend; Step 54 values above remain the historical record.

Executable endpoint replay. A subsequent outcome-blind lock converted only the changed coordinates into immutable SHA-256-prompt/output maps. Runtime requests contain question text and ordered public options but no dataset ID, reference, correctness tag, peer response, pool seed, or selector state. Four endpoints per task reproduce all 12 Step 54 response hashes and all 13 selector-run fields exactly on the same 500 fresh pools. Manifests range from 2,272 to 5,088 bytes; appending a probe character to every supported changed prompt triggers zero overrides. This establishes feasibility under output-only research interfaces, not admission to a particular production registry. The independent validator reports `PASS_STEP55_EXECUTABLE_ENDPOINT_REALISM_AUDIT`.

Calibration-to-holdout, holdout-reference-free audit. A preregistered hash split produces labeled calibration halves of size 500/500/250 and blind holdouts of size 500/500/250 with zero ID overlap. Before holdout outcomes, five-fold calibration selected MiniLM/TF-IDF error learners and froze 1%-distance alias rows; the blind builder received only prompts, public options, and parent outputs. All holdout aliases stay within one accuracy point and at most 1% oracle disagreement. Mean $\Delta CReg$ is $+0.2941$ $[.1079, .4791]$, $+0.1599$ $[.1059, .2141]$, and -0.1124 $[-.1421, -.0835]$ on MedQA, GSM8K, and OpenBookQA. Fixed-query effects are $-0.0306/0/0$. GSM8K selected-support error precision is zero, so this is small-distance approximate control, not exact quality equivalence. The locked machine decision is `PROMOTE_LABEL_FREE_T1`, where “label-free” is scoped only to holdout construction; the validator reports `PASS_STEP56_CALIBRATION_TO_HOLDOUT_AUDIT`.

D NATURAL EXACT-ALIAS AUDIT

The natural anchor is selected by the unique intersection of exact response-and-oracle pairs across MedQA, GSM8K, and OpenBookQA, before running the base-versus-refined selector comparison. The 30-entry stable-order quotient retains frozen index 9; the original 31-entry roster restores index 10 and maps it to root 9. A primary task is path-material at path-change fraction at least .10 and one mean changed position; it is outcome-material at $|\Delta CReg| \geq .1$, a zero-excluding interval, and three-task BH $q < .05$. Across 21 prespecified temperature/budget cells, all cells are path-material and 14 have family-corrected cumulative-regret differences, split into five harms and nine improvements. Candidate-invariant random acquisition gives exact query, root-path, cumulative-regret, and terminal-regret equality in every primary task. The complete Step 50 arrays are stored at `SHA-256 14B77BD3...B0F8EA4`.

A subsequent descriptive replay distinguishes changed positions from unique labels. Mean base-only query counts are 4.358, 1.420, and .960; query sets differ in 94.0%, 87.4%, and 66.0% of runs. Mean first divergence occurs at steps 1.486, 3.698, and 4.690, while mean changed root-decision positions are 10.592, 2.902, and 2.012. The 5% replacement threshold is an analyst-chosen heuristic defined after aggregate Jaccards were known, not confirmatory evidence. Raw arrays and exact random controls are stored in `STEP52_OPERATIONAL_SIGNIFICANCE_RESULTS_2026-08-09.json` (SHA-256 `BB70CEFD...E372FA`).

E CODA OFFICIAL-CODE AUDIT

The official implementation is locked at commit `6c95b7f8`. A prediction-only rule—fixed before labels—selects one parent from the lowest-consensus quartile by the increase in initial family P_{Best} mass after adding four variants. The near variants use $q_j = (1 - \epsilon_j)p + \epsilon_j/C$ for $\epsilon_j \in \{.0005, .001, .0015, .002\}$, preserving every hard prediction and correctness value while changing probability tensors. Exact4 is the multiplicity control. CODA’s EIG configuration and a 30-label budget are fixed; only CoLA activates the public stochastic tie flag, for which all ten frozen seeds are paired.

Table 6: Locked official-code CODA near4 audit. All tasks change their ordered path; exact4 gives the same task-level effects. WNLI ($\dagger$) was not used for target selection.

Pool	Δ cumulative regret	First divergence	Query Jaccard	Δ terminal regret
CoLA	+0.051965	1	0.622	-0.001918
MRPC	+0.053921	5	0.935	0
RTE	+0.841155	2	0.818	0
SST-2	-0.271789	4	0.714	+0.002294
WNLI $\dagger$	-0.014085	1	1.000	0

Near4 changes the ordered path on CoLA, MRPC, RTE, SST-2, and the prospectively unused WNLI pool. Cumulative-regret changes are $+0.051965$, $+0.053921$, $+0.841155$, -0.271789 , -0.014085 , respectively. Exact4 gives the same task-level effects. WNLI has query-set Jaccard one and only reorders its acquisitions. Thus the audit supports cross-method path sensitivity, not response-distinct defense evasion or universal harm.

F PREREGISTERED LLM SELECTOR AUDIT

Outcome-blind freeze and intervention. The official LLM Selector repository is locked at commit 15faa47. Before any judge or weak-judge value was printed, one protocol bound both source-file hashes for all six upstream datasets, included every dataset, and selected the first non-null source-order model as parent. There is no target search. Four aliases inherit the parent’s complete strong-judge outcome row coordinate-wise. Each begins with only the parent’s own ten-weak-judge tensor, then a public SHA-256 rule cycles a frozen 20% of ternary coordinates; it reads no other candidate, strong-judge outcome, pool, acquisition path, or selector state. Realized alias–parent weak-view distances range from 17.25% to 22.25%, while every strong-utility distance is exactly zero. This is a controlled pairwise decision-view refinement over released score matrices, not an executable response wrapper or production-registry submission.

The audit reproduces the official uniform-entry posterior and mean ten-judge expected-entropy acquisition at $(\epsilon_1, \epsilon_2) = (.2, .4)$. For each dataset and seeds 0–999, clean and four-alias registries share a pool of size $\min(400, N)$, budget 30, and deterministic original-root tie rule. Primary terminal regret compares the selected original root with the best original root on that pool. The six-part gate requires coordinate-wise utility identity, exactly zero fixed-query terminal and cumulative effects, path-change rate at least .5, mean terminal harm at least .25 percentage points, positive paired-bootstrap lower bound, and six-task BH-adjusted one-sided sign-flip $q \leq .05$. Two passing datasets were frozen as the promotion threshold; all nulls and improvements remain.

Table 7: Preregistered official LLM Selector audit (1,000 paired pools per dataset). Terminal effects are refined-minus-clean root regret in percentage points. All alias strong-judge rows equal the parent coordinate-wise and every fixed-query effect is zero. A dagger marks the two datasets passing the frozen six-part gate.

Dataset	Δ terminal [95% CI]	BH q	Path $\neq$	Set $\neq$	Root $\neq$	Δ CReg
AlpacaEval [†]	.321 [.194, .453]	.000030	100.0%	99.5%	23.2%	-.245
Arena-Hard	-.370 [-.531, -.213]	1.000	100.0%	99.1%	6.7%	-.379
Bingo [†]	.498 [.374, .623]	.000030	100.0%	98.2%	32.4%	.117
Flickr30k	.183 [-.003, .368]	.05298	100.0%	100.0%	66.7%	-.029
MEDIQA	.000 [.000, .000]	1.000	100.0%	100.0%	.0%	.027
MT-Bench	.000 [.000, .000]	1.000	100.0%	100.0%	.0%	.719

AlpacaEval and Bingo pass, producing the preregistered decision GO_STRONG_CROSS_SELECTOR. The result is directional rather than generic: Arena-Hard significantly improves; Flickr30k’s positive point effect misses both the magnitude and corrected-significance gates; MEDIQA and MT-Bench have terminal effect exactly zero, although their interim cumulative effects are positive. All 6,000 ordered paths change, but query-set and terminal-root changes differ by task. AlpacaEval is especially informative about endpoint choice: terminal regret worsens while cumulative regret improves, so neither metric substitutes for the other.

An independent file that imports no runner verifies all input hashes, reconstructs aliases and every stored pool/statistic, and checks 85,934 invariants. It additionally replays 300 result-independent trajectories from scratch using the literal official entropy equation; all paths, roots, and regrets match. The receipt reports PASS_STEP88_INDEPENDENT_VALIDATION; the preregistration, result, and receipt hashes begin B6303706, 24C3FCDA, and the receipt-bound validator hash 4EEE854C.

G CONFIGURATION-ONLY RAW-INPUT WRAPPER TRANSFER

Outcome separation and endpoint. Step 89 fixed QNLI’s model revision, deterministic calibration/holdout split, complete development grid, one condition, and materiality gate before opening 2,772 sealed outcomes. Four small safetensors files store scalar calibration/configuration values rather than learned neural weights; a CLI accepts only a raw question/sentence pair and emits the parsed hard label with an adapter-specific confidence view. Its 1,000-pool result is negative and remains binding. Step 90 then froze two untouched tasks before any parent prediction: MNLI splits 4,933/4,882 and QQP 20,309/20,121 calibration/holdout items. Per task, all 135 combinations of three roster sizes, three budgets, three temperatures, and five high-confidence trigger fractions were

retained; the top 12 pilot conditions were rerun on 300 calibration pools before one configuration was hashed. Runtime wrappers read only raw text and pinned-parent logits, contain no item map, and exactly preserve hard answers. They establish raw-input execution but are configuration-only response variants, not learned adapters or full checkpoints.

Table 8: Sealed executable calibration-adapter audits (1,000 paired pools each). Terminal effects are refined-minus-clean root regret in percentage points. Hard answers and utilities equal the parent coordinate-wise; every fixed-query effect is zero. MNLI/QQP q -values use Holm correction. QNLI is the retained earlier single-task negative result and reports its raw p .

Task	Trigger	Δ terminal [95% CI]	q/p	Path/final-root change	Gate
QNLI	all confidence views	-.068 [-.149, .014]	.948	96.9/41.7%	No
MNLI	top 30% confidence	+.702 [+.586, .823]	2.0×10^{-5}	99.8/32.8%	Pass
QQP	top 20% confidence	+1.270 [+1.201, +1.340]	2.0×10^{-5}	100.0/82.3%	Pass

MNLI and QQP both pass the preregistered eight-part gate: at least .5 terminal point, positive paired-bootstrap lower endpoint, two-task Holm $q \leq .05$, at least 50% path change, coordinate-wise answer/utility equality, exact fixed-query equality, and matching active-minus-fixed harm. QNLI demonstrates the direction is not guaranteed. A validator importing no runner reconstructs all 4,000 active and 2,000 fixed-query trajectories, both bootstrap/sign-flip analyses, Holm correction, and 12,077 invariants; its receipt is `PASS_STEP90_INDEPENDENT_VALIDATION`. The raw result and validation receipt are hash-bound. This precursor does not satisfy the learned-checkpoint condition addressed next.

H PREREGISTERED LEARNED-ADAPTER DECOUPLED-VIEW AUDIT

New task, outcome separation, and learned parameters. Before downloading an AG News row, Step 92 fixed dataset revision `eb185aad`, base-encoder revision `12040acc`, the full training recipe, split salt, finite attack grid, 1,000 test seeds, estimands, and nine-part pass gate. SHA-256 assigns the 120,000 public training examples to 83,993 root-model, 24,039 adapter-training, and 11,968 attack-calibration items. The complete 7,600-item test split is input-only outside a separately named sealed-label file. A source audit found no earlier AG News result in the project; SST-2 was rejected before new outcomes because it appeared in prior CODA work. This condition uses exact equality of confidence-coded response views for acquisition but hard-label correctness for posterior evidence and regret. It is therefore a decoupled-view selector audit, not an equation-faithful reproduction of Select-LLM’s single similarity.

Twelve bottleneck classification adapters (100,484 learned parameters each) share a frozen DistilBERT encoder and form the clean registry. Each trains without artificial label corruption on a preregistered 2–100% fraction of the root-model partition. For each of five eligible parents, four contextual temperature adapters are independently optimized with four seeds and bootstraps on the disjoint adapter partition. Every saved attack file contains six nonconstant tensors and 50,817 nonzero learned parameters; the four selected files have distinct SHA-256 values. The learned positive temperature rescales parent logits, so the hard label is invariant on every input by construction. A confidence field above a frozen threshold makes the submitted response views distinct. Runtime reads raw text, frozen hidden state, and parent logits—never an item ID, test label, lookup table, peer output, pool, or selector state.

Stage A retains all 300 parent/ τ /bin/trigger configurations on the attack-calibration partition and reruns the top 12 on 300 pools. It freezes parent 8, $\tau = .25$, eight bins, a 30% trigger target, budget 50, and the full 12-root roster. Calibration terminal harm is .609 points with 100% path change. Only then does Stage B open the test labels once.

Table 9: Preregistered learned-adaptor *decoupled-view* audit on 1,000 sealed AG News pools. Confidence-coded responses drive acquisition; hard-label correctness drives posterior evidence and regret. Effects are refined-minus-clean terminal root regret in points; every fixed-query effect is zero.

Task	Params/alias	Pairwise response d	Path/root change	Terminal [95% CI]	AMF [95% CI]	Gate
AG News	50,817	24.1–26.3%	100.0/55.2%	+ .547 [+ .502, + .593]	+ .547 [+ .502, + .593]	Pass

On the complete holdout, realized trigger rates are 30.2–31.5%, pairwise alias-response Hamming distances are 24.1–26.3%, and all 30,400 adapter–item hard answers equal the parent. Ordered paths change in 100% of pools, query sets in 99.3%, and terminal roots in 55.2%. Terminal harm is +.54675 points with paired-bootstrap interval [+ .50225, + .59250] and one-sided sign-flip $p = 9.9999 \times 10^{-6}$; active-minus-fixed has the same mean and a positive interval. Fixed-query root histories and terminal/cumulative effects are byte-identical. The locked decision is GO_SCORE_CHANGING_LEARNED_ADAPTER_BRIDGE.

A validator importing no experimental runner replays all 2,000 active and 2,000 fixed-query trajectories, reconstructs the statistics and gate in 4,170 checks, and runs 256 raw-input end-point samples through all 12 clean checkpoints and four learned adapters. Its receipt is PASS_STEP92_INDEPENDENT_VALIDATION. An initially detected 5.6×10^{-17} acquisition-tie difference motivated a disclosed post-confirmatory audit: sorted, first-appearance, reverse-code, and 10^{-12} tolerance policies change many individual paths but all retain 100% clean–refined path change, exact fixed-query equality, and terminal harm .54575–.54800 points with lower endpoints at least .50125. This robustness audit does not replace the preregistered primary result.

The result closes a learned-parameter, new-task, no-test-lookup bridge for this explicitly decoupled-view selector. It does not establish learned source-faithful Select-LLM terminal harm, independent full-model fine-tuning, live registry admission, secret-task compromise, or deployment prevalence. The next audit tests the single-similarity gap directly.

I PREREGISTERED SOURCE-FAITHFUL LEARNED CONFIRMATION

Before downloading a Banking77 row (Casanueva et al., 2020), Step 93 fixed the processed dataset revision 18072d26, DistilBERT revision 12040acc, hash partitions, 12-root training recipe, seven eligible parents, 504-condition Stage-A grid, 1,000 test seeds, and twelve confirmatory gates. Two pre-outcome amendments record that the pinned executable mirror contains 9,993/3,076 rather than the original metadata’s 10,003/3,080 rows and includes an unused `label_text` column. Both Stage-0 attempts stopped at assertions before row iteration or output; no scientific choice changed. SHA-256 then assigned 6,015/1,996/1,982 training rows to root training, error-predictor training, and calibration. Test texts and labels were separately hash-bound.

Twelve 109,901-parameter intent adapters share a frozen encoder. For each of seven eligible parents, four independently bootstrapped 50,817-parameter networks learn to predict parent error from the hidden vector. A selected alias emits either the parent’s integer label or its unique literal ABSTAIN_a, which is outside all 77 references. Hence its exact-match evidence is coordinate-wise no greater than the parent’s without consulting a test label. The runner accepts response codes and references but no external feedback/utility matrix: the identical function $s(a, b) = \mathbf{1}[a = b]$ constructs candidate-equality groups and reference feedback. The execution lock binds both call sites’ source hashes, all 40 weight files, the complete calibration ledger, sealed labels, and code before output.

Table 10: Preregistered learned single-similarity confirmations on sealed Banking77 (1,000 pools) and selected fresh 20 Newsgroups (2,000 pools). The same exact-match s drives acquisition and posterior evidence. Loss and terminal columns are percentage points; CReg is the 50-step cumulative root-regret difference. Fixed-query effects are exactly zero.

Task	Params/alias	Alias loss	Path/root	Δ Terminal [95%]	Δ CReg [95%]	Gate
Banking77	50,817	.098–.358	99.8/36.6%	+ .032 [– .055, + .122]	+ .0409 [+ .0129, + .0689]	No
20 Newsgroups	205,245	.252–.491	100/48.2%	+ .120 [– .044, + .290]	+ .0822 [+ .0385, + .1266]	No

The selected parent is root 10 with $\tau = .1$, budget 50, and a .25-point calibration-loss cap. On the 3,076-item holdout, aliases lose .098–.358 accuracy points, trigger on .10–.52% of items, and are coordinate-wise non-improving. Across 1,000 paired pools, 99.8% of ordered paths, 99.1% of query sets, and 36.6% of terminal roots change. Cumulative regret rises by +.040945 [+ .012905, +.068910], but terminal regret rises only +.032 [−.055, +.122] percentage points with one-sided $p = .236$. Active-minus-fixed terminal harm is identical; all fixed-query root histories and terminal/cumulative effects are exactly equal. The preregistered .5-point terminal gate therefore fails and the retained decision is NO_GO_RETAIN_SOURCE_FAITHFUL_NEGATIVE.

The first machine result printed INVALID_STEP93_IMPLEMENTATION because a supplemental dense-versus-sparse acquisition check required absolute error at most 2.0×10^{-16} ; the two algebraically equal reductions differed by 2.220446×10^{-16} , one binary64 ULP at one. Source hashes, bitwise reference feedback, endpoint replay, coordinate constraints, trajectories, and all scientific gates were unaffected. An independent implementation reconstructs all 1,000 clean/refined active and fixed runs, inference, and 128 raw-input endpoints in 82 checks and adjudicates the valid retained negative without promoting any gate or changing an outcome. No replacement task is opened.

J FRESH-TASK LEARNED SINGLE-SIMILARITY REPLICATION

After retaining the Banking77 terminal negative, Step 94 fixed a finite menu—20 Newsgroups, MASSIVE en-US intent classification, and SST-5—before materializing their test splits. SHA-256 partitions each development population into root training, gate training, selector search, and selector verification. Sixteen independently trained bottleneck classifiers per task form a candidate bank. Three prespecified quality-gap templates deterministically select eight-root rosters from search accuracy; only the top three search roots are eligible parents. Four independently bootstrapped error gates per eligible parent consume frozen CLS vectors and parent logits. An integrated variant checkpoint contains a copy of the parent classifier plus one learned gate and accepts raw text only. Above its frozen threshold it emits a unique out-of-alphabet ABSTAIN_a; otherwise it emits the parent class, guaranteeing coordinate-wise non-improvement without test outcomes.

The complete search comprises 648 conditions per task. The top 24 are rerun on 500 disjoint development-verification pools, after which a locked rule selects the task with largest positive active-minus-fixed terminal effect among eligible tasks. All three are eligible; verification effects are +.727/+.208/+.134 terminal points for 20 Newsgroups/MASSIVE/SST-5, with 100% path change and exact fixed-query equality. The rule selects 20 Newsgroups, root 10, the eight-root roster (10, 9, 6, 3, 1, 0, 2, 4), $\tau = .25$, and four thresholds. A lock binds the complete 1,944-cell search ledger, 72 verification rows, 128 learned files, code, the selected test, and 2,000 confirmatory seeds before only the selected test label file is opened. The other two test outcomes remain unanalyzed.

Each selected integrated checkpoint has 205,245 nonzero learned parameters and a distinct SHA-256 digest. Raw-input replay exactly reproduces the parent component and never accepts a label, item ID, lookup, peer output, pool, trajectory, posterior, or selector state. On 7,532 sealed test items, parent accuracy is 60.847%; aliases lose .252–.491 points and trigger on 1.53–2.20%. Every ordered path and query set changes, as do 48.2% of terminal roots. Cumulative regret increases by +.082157 [+ .038544, +.126554] with one-sided sign-flip $p = .000200$. Terminal regret increases by only +.1198 [−.0444, +.2895] percentage points with $p = .0813$, below the frozen .5-point gate; active-minus-fixed is identical. Fixed-query root histories and both regret arrays are bitwise equal. The retained decision is NO_GO_RETAIN_FRESH_TASK_NEGATIVE: a second source-faithful learned cumulative result, not a terminal success.

An independent validator reads the locked raw arrays rather than the experimental runner and reconstructs hash bindings, all delta vectors, coordinate non-improvement, quality losses, bootstrap intervals, sign-flip values, path changes, fixed-query equality, every gate, and the retained decision. All checks pass. This audit strengthens replication and raw-endpoint realism for the paper’s primary anytime endpoint, but does not close the stronger independently fine-tuned full-checkpoint or material terminal-harm condition.

K PROSPECTIVE UNTARGETED-REFINEMENT STRESS TEST

After the targeted experiments were complete, we prospectively locked 14 previously unused tasks: seven official MODEL SELECTOR tasks and seven HELM MATH subjects. The audit crossed exact, near, and independent-diverse additions, producing 420 nonclean task-condition units with 32 paired runs each. All response-only scores and a 50% high-risk set were sealed before outcome access. Only 35/420 effects crossed the preregistered materiality threshold: one beneficial MODEL SELECTOR effect and 34 Select-LLM effects, of which eight were harmful and 26 beneficial. Exact quotienting neutralized all exact additions, and authenticated lineage neutralized every registered exact and near addition by construction.

The attempted ex-ante registry-fragility audit did not transfer. Its magnitude Spearman correlation was .185 with task-block permutation $p = .416$; AUROC was .719, below a static mass-displacement baseline at .788; and it failed the magnitude, materiality, adaptive-value, cross-selector, and fallback gates. Only the non-vacuity gate passed. The locked decision is therefore `STOP_FRAGILITY_PIVOT`. These results rule out a generic “more entries are harmful” interpretation and any claim that our response-only score predicts attacks. They do not test the separately locked targeted T2-A constructor, whose support is deliberately chosen rather than random. The full protocol, 420 effects, six gates, and 98-check independent validation are stored in the Step 61 artifacts.

L DEFENSE AND MUTATION AUDIT

Table 11: Matched 10% behavioral-cover audit. Clean Δ compares covered clean with uncovered clean; positive is worse. “Extra rm.” counts removals beyond exact quotienting.

Task	Undefended Δ CReg [95% CI]	Cover residual	Clean Δ CReg	Extra rm.	Max span
MedQA	+4.853 [4.603, 5.102]	0	+0.569	1	0.8 pp
OpenBookQA	+1.006 [0.942, 1.069]	0	+1.030	6	5.2 pp
GSM8K	Prospectively ineligible; frozen parent not replaced				

Table 12: Representative post-hoc fixed settings. R is absolute residual CReg divided by the undefended effect over five attacks; Clean G/M/O is Δ CReg (positive is worse). All 21 settings appear in the appendix.

Rule, α	Mean R	Max R	$R < 1$	Clean G	Clean M	Clean O
All three, .050	.824	1.061	3/5	−0.180	+0.351	+0.238
MCCA, .100	.765	1.121	3/5	−0.180	+0.275	+1.018
MCCP, .100	.721	1.125	4/5	−0.180	+0.275	+1.112
class-unif., .132	.656	1.185	4/5	−0.180	+0.386	+1.110
MCCA, .132	.584	1.158	4/5	−0.180	+0.386	+1.273

The exact quotient groups identical full response rows and exactly neutralizes the natural pair and T0 copies. A radius- ρ matched cover joins candidates within normalized response Hamming distance ρ and retains the smallest-index representative of each connected component. At 5%, it neutralizes the corresponding attack with incremental clean CReg 0/0/ −.169 across MedQA/GSM8K/OpenBookQA. At 10%, it also neutralizes the attack, but clean CReg increases by .569 on MedQA and 1.030 on OpenBookQA; the latter merges six natural models spanning 5.2 accuracy points and costs 3.975 terminal points. Authenticated lineage returns exact equality by retaining fixed root mass, but is explicitly an oracle-information comparator.

The soft-weight audit crosses class-uniform, MCCA, and MCCP with $\alpha \in \{.01, .05, .10, .132, .15, .184, .213\}$. Its 63 clean and 105 eligible attack cells are stored without outcome filtering in `STEP46_CLONE_ROBUST_SOFT_WEIGHTING_RESULTS.2026-08-08.json` (SHA-256 15B693F0...B7B136). The original joint regret/path gate remains 0/105 and all residual path-change fractions equal one. The outcome-first reanalysis instead finds 74 mitigated and 31 worsened attack cells; 24 reach at most 50% residual, 11 at most 25%, and four at most 10%. No fixed setting is non-worsening on all five task-distance attacks. All 21 settings improve clean

GSM8K CReg but worsen clean MedQA and OpenBookQA CReg; terminal intervals contain zero in 45/63 cells, with ten significant harms and eight improvements. Table 13 reports every fixed setting. Independent validators reconstruct graph breakpoints, priors, paths, effects, and mutation failures rather than trusting recorded decisions.

Table 13: Complete fixed-configuration soft-weight audit. Mean and maximum R summarize absolute residual CReg ratios over five eligible attack cells; “mit.” counts $R < 1$. Clean columns report Δ CReg on GSM8K/MedQA/OpenBookQA. Final is the largest absolute clean terminal shift across the three tasks, in percentage points.

Rule	α	Mean R	Max R	Mit.	Clean G	Clean M	Clean O	Final pp
class-unif.	.010	.814	1.061	3/5	−.180	+.351	+.338	.023
class-unif.	.050	.824	1.061	3/5	−.180	+.351	+.238	.023
class-unif.	.100	.830	1.153	3/5	−.180	+.275	+.973	.020
class-unif.	.132	.656	1.185	4/5	−.180	+.386	+1.110	.203
class-unif.	.150	.644	1.400	4/5	−.314	+.555	+1.108	.181
class-unif.	.184	.744	1.307	4/5	−.211	+.609	+1.455	.037
class-unif.	.213	.730	1.550	4/5	−.160	+.638	+1.290	.037
MCCA	.010	.814	1.061	3/5	−.180	+.351	+.338	.023
MCCA	.050	.824	1.061	3/5	−.180	+.351	+.238	.023
MCCA	.100	.765	1.121	3/5	−.180	+.275	+1.018	.022
MCCA	.132	.584	1.158	4/5	−.180	+.386	+1.273	.024
MCCA	.150	.637	1.295	4/5	−.314	+.555	+1.172	.037
MCCA	.184	.597	1.259	4/5	−.211	+.609	+2.010	.037
MCCA	.213	.714	1.497	4/5	−.160	+.638	+2.483	.037
MCCP	.010	.814	1.061	3/5	−.180	+.351	+.338	.023
MCCP	.050	.824	1.061	3/5	−.180	+.351	+.238	.023
MCCP	.100	.721	1.125	4/5	−.180	+.275	+1.112	.175
MCCP	.132	.734	1.162	4/5	−.180	+.386	+1.897	.181
MCCP	.150	.785	1.390	4/5	−.314	+.555	+1.823	.169
MCCP	.184	.874	1.304	3/5	−.211	+.609	+2.933	.037
MCCP	.213	1.064	1.584	3/5	−.160	+.638	+3.724	.037

M ARTIFACT MAP

The artifact root maps every main-text number to an immutable result and an independent validator. The primary families are Step 27 (formal checker), Steps 30–31 (complete T2-M grid and covers), Step 35 (official CODA), Steps 46–49 (soft weights), Steps 50–52 (natural pair), Steps 54–55 (peer-free executable T2-A), Step 56 (within-task T1), Step 61 (prospective untargeted stress), Step 80 (canonical V2 transfer), Step 87 (prospective fresh-task T1), Step 88 (official LLM Selector terminal audit), Steps 89–90 (sealed configuration-only response wrappers), Step 92 (learned decoupled-view AG News), Step 93 (learned source-faithful Banking77), and Step 94 (fresh-task learned single- s replication). Each contains a protocol or preregistration, execution lock, raw result, and validation output. The anonymous manifest binds the revised paper plus Step 87’s outcome separation, Step 88’s six-dataset arrays, Step 90’s two positive and one retained negative wrapper audits, Step 92’s learned positive result and numeric-tie audit, Step 93’s 40 learned weights and retained negative, and Step 94’s complete three-task development ledger, 128 learned files, one selected sealed result, and independent reconstruction. Offline validation checks every bundled byte and anonymity. Top-level full validation verifies inherited closure receipts and independently reconstructs the current learned single- s audits; earlier-stage validators remain separate entry points. Public source packages, base weights, and third-party repositories are not embedded; this is validation closure, not regeneration of exploratory history.